

ZYRYANKA, Russian Federation

WMO: 254000

Lat: 65.7468N

Lon: 150.8870E

Elev: 43

StdP: 100.81

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -48.0	(c) -46.4	(d) -49.5	(e) 0.0	(f) -47.8	(g) -47.8	(h) 0.0	(i) -46.3	(j) 7.0	(k) -25.2	(l) 6.2	(m) -26.2	(n) 0.4	(o) 100	(p) 0.785

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.0	(c) 27.8	(d) 19.0	(e) 25.4	(f) 17.6	(g) 23.1	(h) 16.3	(i) 19.9	(j) 26.2	(k) 18.4	(l) 23.9	(m) 17.0	(n) 22.3	(o) 3.1	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	12.6	23.5	16.0	11.5	21.6	14.6	10.4	20.2	57.3	26.5	52.5	23.9	48.2	22.3	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.7	6.5	5.7	-49.5	31.7	1.6	2.3	-50.7	33.3	-51.7	34.6	-52.6	35.9	-53.8	37.6
			-49.6	21.6	1.6	1.9	-50.7	22.9	-51.7	24.0	-52.6	25.1	-53.7	26.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.9	-36.1	-33.3	-23.3	-10.4	3.8	13.7	16.1	12.4	4.9	-9.4	-25.0	-33.8
	DBStd	19.95	6.66	6.34	6.67	6.66	5.84	4.66	4.76	4.35	4.34	6.18	7.17	7.16
	HDD10.0	7707	1428	1212	1031	611	206	18	5	25	162	601	1051	1358
	HDD18.3	10349	1686	1445	1289	861	451	148	103	189	402	859	1301	1616
	CDD10.0	448	0	0	0	0	14	131	193	99	10	0	0	0
	CDD18.3	48	0	0	0	0	0	11	33	5	0	0	0	0
	CDH23.3	463	0	0	0	0	2	111	303	47	1	0	0	0
	CDH26.7	118	0	0	0	0	0	19	92	7	0	0	0	0
Wind	WSAvg	2.3	1.2	1.6	2.2	2.4	3.0	3.1	2.8	2.7	2.7	2.3	1.9	1.6
Precipitation	PrecAvg	291	18	15	11	8	15	34	50	44	32	23	23	17
	PrecMax	482	50	44	31	38	43	80	177	108	123	111	57	40
	PrecMin	144	2	1	1	0	0	4	5	5	0	3	4	2
	PrecStd	78	11	9	8	7	10	18	34	25	26	17	13	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-17.2	-16.3	-2.3	7.8	22.5	29.0	32.1	27.2	21.1	5.9	-6.3	-14.4
		MCWB	-17.5	-16.9	-4.9	3.1	13.8	19.3	21.3	18.6	13.5	3.0	-7.1	-14.9
	2%	DB	-20.8	-19.7	-5.5	5.2	18.9	26.4	29.2	24.4	17.0	1.9	-10.1	-18.3
		MCWB	-21.1	-20.0	-7.4	1.3	11.6	17.1	20.2	17.1	11.0	-0.3	-10.5	-18.8
	5%	DB	-23.8	-21.8	-8.9	2.9	16.0	24.1	26.7	22.3	14.3	0.0	-13.1	-21.2
		MCWB	-24.0	-22.1	-10.2	-0.2	9.8	15.8	18.8	16.0	9.8	-1.1	-13.6	-21.4
	10%	DB	-26.2	-23.7	-13.0	0.5	13.3	21.8	24.2	19.9	11.7	-1.4	-15.4	-23.2
		MCWB	-26.4	-23.9	-13.8	-1.9	8.1	14.6	17.7	14.9	8.3	-2.5	-15.9	-23.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-17.5	-16.6	-4.8	3.2	14.3	19.9	22.6	19.6	13.7	2.9	-7.0	-14.9
		MCDB	-17.2	-16.2	-2.4	7.3	21.8	28.1	30.5	26.2	19.8	5.4	-6.4	-14.6
	2%	WB	-21.1	-20.0	-7.3	1.4	12.0	17.9	21.0	17.8	11.7	0.1	-10.5	-18.7
		MCDB	-20.8	-19.8	-5.5	5.0	18.3	25.2	27.6	23.1	15.8	1.2	-10.1	-18.3
	5%	WB	-24.0	-22.0	-10.2	-0.1	10.1	16.4	19.5	16.5	10.0	-1.1	-13.5	-21.5
		MCDB	-23.8	-21.8	-8.9	2.8	15.4	22.8	25.2	21.1	13.8	0.1	-13.1	-21.3
	10%	WB	-26.4	-23.9	-13.8	-1.8	8.2	15.1	18.2	15.2	8.5	-2.5	-15.8	-23.4
		MCDB	-26.2	-23.7	-13.0	0.5	13.2	21.0	23.2	19.4	11.5	-1.3	-15.5	-23.2
Mean Daily Temperature Range	5% DB	MDBR	6.0	8.0	11.8	12.2	9.2	9.5	9.0	8.4	6.8	5.7	5.9	5.8
		MCDBR	6.6	6.8	15.5	11.9	12.8	13.1	13.3	12.7	12.4	6.9	6.4	6.8
		MCWBR	6.4	6.7	13.7	9.0	7.4	6.3	6.0	6.4	7.1	5.0	6.2	6.6
	5% WB	MCDBR	6.5	7.1	15.4	11.4	12.4	12.5	11.9	11.3	11.4	6.6	6.6	6.7
		MCWBR	6.4	6.9	13.6	8.7	7.2	6.5	5.9	6.2	6.7	4.8	6.4	6.5
Clear-Sky Solar Irradiance	taub	0.170	0.239	0.297	0.351	0.392	0.383	0.497	0.450	0.363	0.272	0.182	0.174	
	taud	1.909	2.081	2.058	1.992	2.154	2.369	2.003	2.097	2.282	2.207	1.919	1.903	
	Ebn at Noon	310	635	751	793	802	827	696	690	684	582	260	53	
	Edh at Noon	19	56	96	133	126	105	148	122	80	49	17	5	
All-Sky Solar Radiation	RadAvg	0.11	0.80	2.59	4.96	5.74	5.68	4.93	3.66	2.12	0.89	0.21	0.02	
	RadStd	0.01	0.06	0.09	0.14	0.32	0.35	0.51	0.35	0.31	0.11	0.02	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.69	+1.18	+1.30	N/A	N/A	N/A	-301	-290	N/A	-23
Regional (1 neighbor)	+0.69	+1.18	+1.30	N/A	N/A	N/A	-301	-290	N/A	-23

Nomenclature: See separate page